

Applying ODE-Based Pharmacokinetic Models to Optimize Antihistamine Dosing Schedules

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Abstract: This paper examines ODE modelling methods to optimize antihistamine dosing schedules based on individual information. Pharmacokinetics and pharmacodynamics are used to determine drug doses, however these practices only reflect the general public, and not those at the extremes. Both three-compartment and four-compartment models are used to model the pharmacodynamics of antihistamines, through the use of both Heun's Method and the Runge-Kutta (4) method.

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1 Introduction

1.1 Process of Drug Dosing

The field of study regarding dosing schedules is currently dictated by pharmacokinetics and pharmacodynamics. The former concerns the actions of drugs on the body, including the four stages: absorption, distribution, metabolism and excretion of drugs [1]; the latter concerns the drugs' actions in the patient's body [1].

1.1.1 Pharmaocokinetics

Pharmacokinetics is crucial for determining the drug's dosage and safety, and acts as the basis for compartment models of various drugs [2].

Absorption: The process of administering the drug at a particular site, the site varies by medication type and medical methods. In this report, the oral route under normal circumstances [3], and the intravenous route [4] is further investigated.

Distribution: The process of drug dispersal throughout the body. The distribution rate is related to molecular structure and tissue type [5, 6].

Metabolism: The process where the drug reaches the liver and is metabolized, causing desirable effects by reactions (Phase-I). Conjugation (Phase-II) then occurs to ready the excretion of the drug [1, 7].

Excretion: The process of eliminating the drugs from the body, this is typically done through the renal system [1], and thus is variable due to kidney function. Another variable is the drug's half-life – the time it takes for the concentration of the drug to halve in the body. The half-life depends on drug properties and the person subject to the drug [8].

1.1.2 Pharmacodynamics

Pharmacodynamics, based on Receptor Theory [9, 10], classifies chemical reactions triggered by drugs into direct/indirect and immediate/delayed effects. It provides the following information on drug modelling [11]:

- Toxicity of the drug and hence the therapeutic window of the drug [12].
- Receptor-drug relation and receptor quantity determine the magnitude of the response [11].
- Relation to drug distribution based on drug type leading to the use of the Compartmental Model.

1.2 Drug Delivery for Antihistamines

Antihistamines are antagonistic drugs, meaning they bind to and block histamine receptors, which release histamine [13, 14]. Histamine is a chemical with the following effects: dilating blood vessels, inflammation, changes in muscle contraction and heart rate, and causes allergy symptoms including nasal congestion, sneezing, and hives.

Histamine receptors can be separated into two types: H1 receptors, which are associated with allergy symptoms, and H2 receptors, which affect the gastrointestinal tract. To model and compare three kinds of antihistamines, only second generation H1 antihistamines will be considered, specifically cetirizine, fexofenadine, and loratadine.

Second generation H1 antihistamines bind to the H1 receptors in the body, and specifically decrease vascular permeability, and demonstrate anti-inflammatory properties [15]. While the

three antihistamines being compared all have this mechanism of action, they do differ in some key ways. Loratadine has the fastest onset of action, with the shortest half-life, with its primary metabolism being in the liver [16]. Fexofenadine inhibits mast cell and inflammatory cell release, and demonstrates a higher inhibition rate [17]. Cetirizine demonstrated the best anti-inflammatory activity, and targets allergic rhinitis by inhibiting histamine induced bronchospasms.

1.3 Motivation

Currently, a drug’s dose schedule reflects the “general public”, which excludes those at the extremes of age, weight, as well as people who are pregnant, and those with genetic disorders [18]. Major factors that contribute to varying absorption schedules include these aforementioned attributes [19] and vary by absorption method [20].

Our group chose antihistamines, as allergies are present in our daily life. Furthermore, the widespread use and commercialization of antihistamine products make it a relevant topic of interest. This motivates the research question: “How can ODE-based pharmacokinetic models be used to optimize antihistamine dosing schedules for individuals?”

Drug dosing schedules are a suitable topic for modelling due to the following:

1. Little modelling has been developed within the field, and in most cases, systems of ODEs with only a few parameters are applied, including compartmental models [21], and PBPK models [22].
2. Both health [23] and ethical [24] concerns imply clinical trials are inappropriate for this field.
3. Factors affecting dosing schedules such as age, gender, and weight are suitable factors to parameterize and model ODEs.

2 Mathematical Model and Theoretical Foundation

Pharmacokinetics can be described by multi-compartmental ordinary differential equations [25] by treating different spaces in the body as compartments and simplifying predictions of instantaneous drug concentration as they enter and leave each one. The rate at which the drug flows is proportional to the concentration of drug present in the first compartment, where the constant of proportionality is determined by the drug, compartment, and health of individual [26]. The rate of change in concentration at a given time can then be denoted by:

$$\frac{dC}{dt} = \text{input rate of drug} - \text{output rate of drug} \quad (1)$$

2.1 Modeling Assumptions

1. **Within the body, the drug is equally soluble in all body tissues:** Simplifies the mathematical framework to depend only on concentrations.
2. **At the reaction site, the molecule is consumed by the body and can no longer affect the system:** This is a mathematical simplification of the process of metabolism and elimination, allowing a linear elimination rate related to concentration.
3. **All of the body can be modelled as the reaction site, the storage, generally fat, or blood, with anything else having no effects:** This allows the simplification of the body into compartments, ignoring tissues that are of low permeability [6].

4. **The drug is immediately able to evenly propagate through the entire bloodstream at once:** Since the distribution phase is short [27], this assumption removes the spatial dependence of the differential equations, allowing it to simplify into an ODE dependent on concentrations.

3 Intravenous Administration

3.1 IV Three Compartment Model

Three-compartment models are widely used as they account for the rapid initial distribution phase, and the subsequent slower elimination phase sustained by the gradual release of the drug from tissue storage sites [28]. The model considered three interconnected compartments: blood (C_b), fat (C_f), and the reaction site (C_r), where the concentration vector

$$C = \begin{bmatrix} C_b(t) \\ C_f(t) \\ C_r(t) \end{bmatrix}, \quad C(0) = \begin{bmatrix} C_0 \\ 0 \\ 0 \end{bmatrix} \quad (2)$$

assuming that the drug is initially introduced only into the blood compartment.

3.2 Three-Compartment Parameters

The parameters represent first-order rate constants that govern the flow between compartments and elimination phases:

Reversible blood-fat exchange	k_{bf}
Reversible blood-reaction exchange	k_{br}
Irreversible elimination from blood	k_{bl}
Irreversible absorption from reaction site	k_{rl}

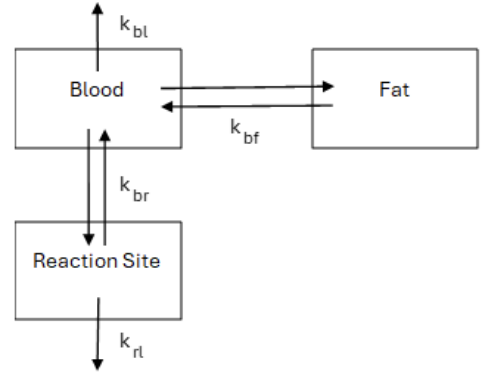


Figure 1: 3-compartment flow

3.3 System of Ordinary Differential Equations

The change in concentration over time is modeled as a linear system:

$$\frac{dC}{dt} = AC \quad (3)$$

where A is the derivative matrix:

$$A = \begin{bmatrix} -(k_{bl} + k_{bf} + k_{br}) & k_{bf} & k_{br} \\ k_{bf} & -k_{bf} & 0 \\ k_{br} & 0 & -(k_{bl} + k_{rl}) \end{bmatrix}, \quad C(0) = \begin{bmatrix} -(k_{bl} + k_{bf} + k_{br})C_0 \\ k_{bf}C_0 \\ k_{br}C_0 \end{bmatrix} \quad (4)$$

This matrix equation expands into the following three coupled linear differential equations:

$$f(C_b, C_f, C_r) = \frac{dC_b}{dt} = -(k_{bl} + k_{bf} + k_{br})C_b + k_{bf}C_f + k_{br}C_r \quad (5)$$

$$g(C_b, C_f) = \frac{dC_f}{dt} = k_{bf}C_b - k_{bf}C_f \quad (6)$$

$$h(C_b, C_r) = \frac{dC_r}{dt} = k_{br}C_b - (k_{br} + k_{rl})C_r \quad (7)$$

3.4 Computational Approach

3.4.1 Heun's Method

In our 3-variable time-independent model, the value in the next time step using the Improved Euler's Method can be expressed with 3 variables C_b, C_f, C_r , as:

$$C_{b,n+1} = \frac{[f(\widetilde{C_{b,n+1}}, \widetilde{C_{f,n+1}}, \widetilde{C_{r,n+1}}) + f(C_{b,n}, C_{f,n}, C_{r,n})]\Delta t}{2} + C_{b,n} \quad (8)$$

$$\text{where } \widetilde{C_{b,n+1}} = f(C_b, C_f, C_r)\Delta t + C_{b,n}$$

Only C_b Equations are shown here, see Appendix 8.1 for full Equations.

This method produces okay results; the local and global truncation error produced is of order $O(\Delta t^3)$ and $O(\Delta t^2)$ respectively.

3.4.2 Runge-Kutta Method

The other method our group attempted was the Runge-Kutta (4) Method, which can be expressed as the following slope values, and the value of the next time step is given by:

Initial slopes:

$$F_1 = f(C_{b,n}, C_{f,n}, C_{r,n}) \quad (9)$$

Halfway slopes:

$$F_2 = f(C_{b,n} + \frac{\Delta t}{2}F_1, C_{f,n} + \frac{\Delta t}{2}G_1, C_{r,n} + \frac{\Delta t}{2}H_1) \quad (10)$$

Readjusted Halfway slopes:

$$F_3 = f(C_{b,n} + \frac{\Delta t}{2}F_2, C_{f,n} + \frac{\Delta t}{2}G_2, C_{r,n} + \frac{\Delta t}{2}H_2) \quad (11)$$

End slopes:

$$F_4 = f(C_{b,n} + \Delta t F_3, C_{f,n} + \Delta t G_3, C_{r,n} + \Delta t H_3) \quad (12)$$

Then the value at the next time step is given by taking a weighted average of different slopes,

$$C_{b,n+1} = C_{b,n} + \frac{\Delta t}{6}(F_1 + 2F_2 + 2F_3 + F_4), \quad (13)$$

Only C_b Equations are shown here, see Appendix 8.2 for full Equations.

This method produces significantly improved errors while having a reasonable computational time; local and global truncation error is of order $O(\Delta t^5)$ and $O(\Delta t^4)$ respectively.

4 Oral Administration

4.1 Oral Four Compartment Model

The four-compartment pharmacokinetic model extends the conventional three-compartment model by adding an initial absorption compartment to simulate the pharmacokinetics of an orally administered drug. It is assumed that the drug is initially introduced only in the digestive system (C_d), then moving into the blood (absorption), followed by distribution and elimination from the blood and peripheral compartments (reaction site and fat). This structure includes the absorption phase required for oral dosing, along with the central and two peripheral storage sites, providing a complete representation of drug movement.

4.2 Four-Compartment Parameters

The parameters represent first-order rate constants that govern the flow between compartments and elimination phases:

Irreversible digestive system-blood exchange	k_{db}
Reversible blood-reaction exchange	k_{br}
Reversible blood-fat exchange	k_{bf}
Irreversible elimination from blood	k_{bl}
Irreversible absorption from digestive system	k_{dl}
Irreversible absorption from reaction site	k_{rl}

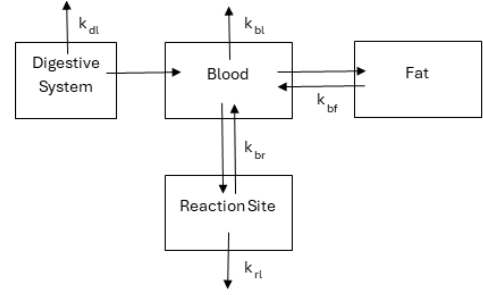


Figure 2: 4-compartment flow

4.3 System of Ordinary Differential Equations

The change in concentration over time is modeled the same as for intravenous administration (equation (3)), where

$$A = \begin{bmatrix} -(k_{db} + k_{bl}) & 0 & 0 & 0 \\ k_{db} & -(k_{bl} + k_{bf} + k_{br}) & k_{bf} & k_{br} \\ 0 & k_{bf} & -k_{bf} & 0 \\ 0 & k_{br} & 0 & -(k_{br} + k_{rl}) \end{bmatrix} \quad (14)$$

This matrix equation expands into the following three coupled linear differential equations:

$$f(C_b, C_f, C_r, C_d) = \frac{dC_b}{dt} = k_{db}C_d - (k_{bl} + k_{bf} + k_{br})C_b + k_{bf}C_f + k_{br}C_r \quad (15)$$

$$g(C_b, C_f) = \frac{dC_f}{dt} = k_{bf}C_b - k_{bf}C_f \quad (16)$$

$$h(C_b, C_r) = \frac{dC_r}{dt} = k_{br}C_b - (k_{bf} + k_{rl})C_r \quad (17)$$

$$m(C_d) = \frac{dC_d}{dt} = -(k_{db} + k_{dl})C_d \quad (18)$$

4.4 Computational Approach

4.4.1 Heun's Method

The improved Euler's method was used here, again equations are time independent with 4 variables C_b, C_f, C_r, C_d :

$$C_{b,n+1} = \frac{[f(\widetilde{C_{b,n+1}}, \widetilde{C_{f,n+1}}, \widetilde{C_{r,n+1}}, \widetilde{C_{d,n+1}}) + f(C_{b,n}, C_{f,n}, C_{r,n}, C_{d,n})]\Delta t}{2} + C_{b,n} \quad (19)$$

$$\text{where } \widetilde{C_{b,n+1}} = f(C_b, C_f, C_r, C_d)\Delta t + C_{b,n}$$

Only C_b Equations are shown here, see Appendix 8.3 for full Equations.

Local and global truncation error produced is of order $O(\Delta t^3)$ and $O(\Delta t^2)$ respectively.

4.4.2 Runge-Kutta Method

Applying the Runge-Kutta (4) Method, the following slope values are obtained:

Initial slopes:

$$F_1 = f(C_{b,n}, C_{f,n}, C_{r,n}, C_{d,n}), \quad (20)$$

Halfway slopes:

$$F_2 = f\left(C_{b,n} + \frac{\Delta t}{2}F_1, C_{f,n} + \frac{\Delta t}{2}G_1, C_{r,n} + \frac{\Delta t}{2}H_1, C_{d,n} + \frac{\Delta t}{2}M_1\right) \quad (21)$$

Readjusted Halfway slopes:

$$F_3 = f\left(C_{b,n} + \frac{\Delta t}{2}F_2, C_{f,n} + \frac{\Delta t}{2}G_2, C_{r,n} + \frac{\Delta t}{2}H_2, C_{d,n} + \frac{\Delta t}{2}M_2\right) \quad (22)$$

End slopes:

$$F_4 = f(C_{b,n} + \Delta t F_3, C_{f,n} + \Delta t G_3, C_{r,n} + \Delta t H_3, C_{d,n} + \Delta t M_3) \quad (23)$$

Then the value at the next time step is given by taking a weighted average of different slopes,

$$C_{b,n+1} = C_{b,n} + \frac{\Delta t}{6}(F_1 + 2F_2 + 2F_3 + F_4) \quad (24)$$

Only C_b Equations are shown here, see Appendix 8.4 for full Equations.

The local and global truncation error is of order $O(\Delta t^5)$ and $O(\Delta t^4)$ respectively.

5 Analysis of Model

To model the body, three levels of complexity were used. The two-compartment model was a simplified way to provide a competent understanding of the system, and ensure the graphing worked. However, the limitations of having a relatively rapid decay rate due to a lack of storage element provided issues. It meant the reaction absorption had to be much slower and blood losses had to be significantly lower, thus leading to a new model being required: The three-compartment model, as seen in Figure 3. More specifically, the model reveals how the concentration of the antihistamine changes in both the bloodstream and storage in the body.

The three-compartment model was a better approximation, but treated the drug as though it was administered intravenously, whereas many antihistamines are orally administered.

Runge-Kutta method approximation of three compartment concentrations with $N = 400$

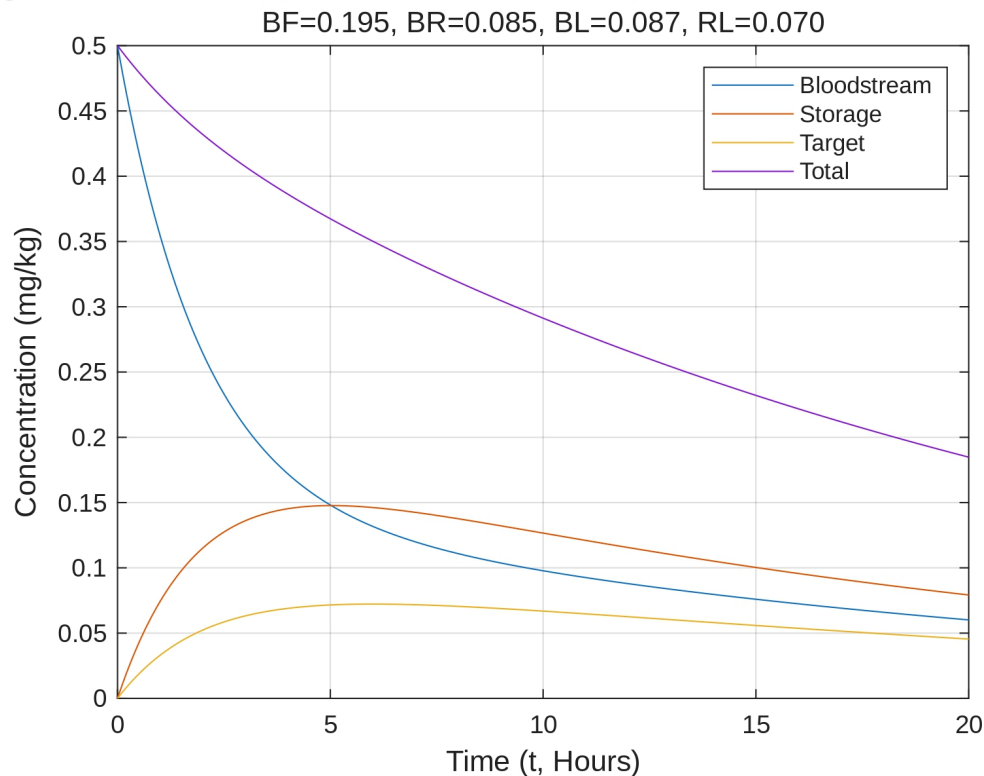


Figure 3: *Three-compartment model demonstrating how the concentration of antihistamine in the body varies with time.*

This led to the use of the four-compartment model, as seen in Figure 4. The graph reveals how the concentrations of antihistamines change in the body, and includes the impact of the digestive system. This inclusion is critical in developing an accurate model, as oral ingestion is the most common way antihistamines are administered, and thus needs to be included.

Because of a lack of information pertaining to how each drug reacts, the parameters were determined on the basis of length between doses, or the half-life of the drug in the bloodstream from the peak. This determination gave a set of experimental coefficients of interaction, which simulated the effectiveness of transmission to different body compartments, while using the assumption of equal solubility.

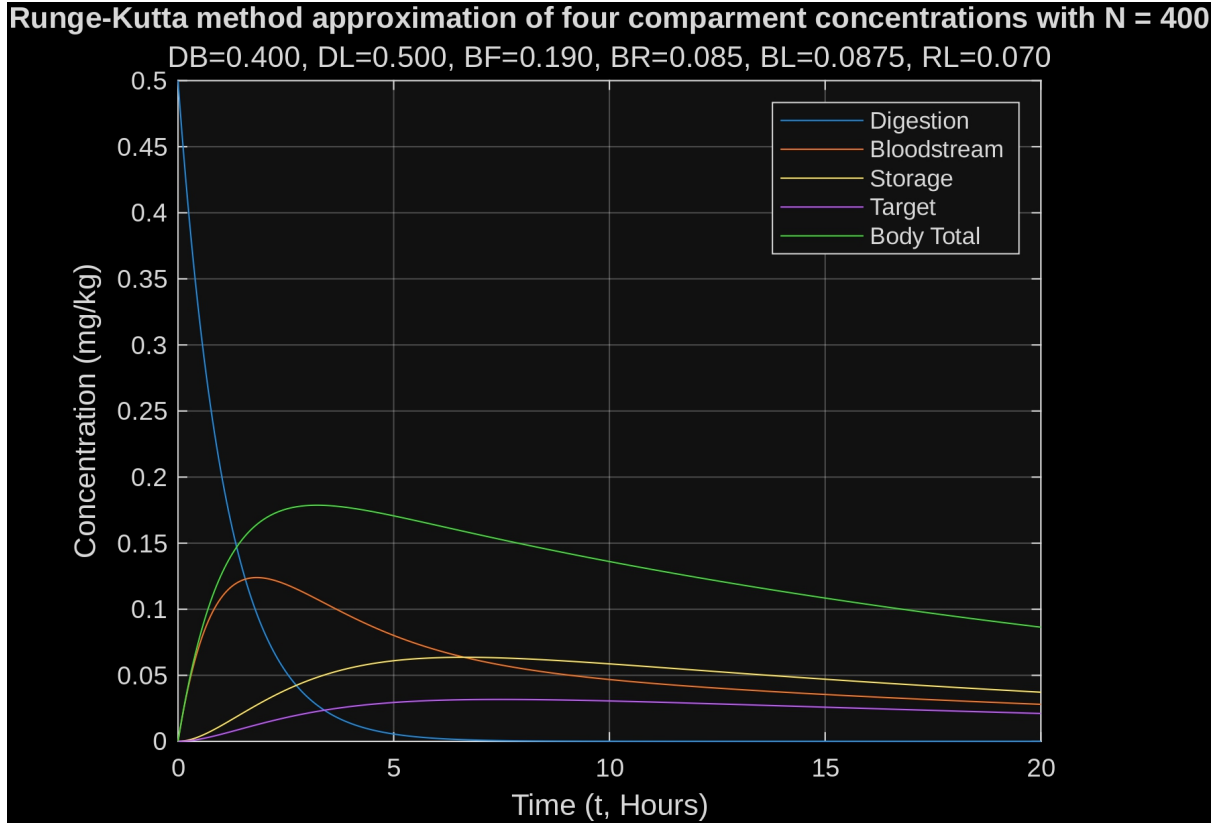


Figure 4: *Four-compartment model demonstrating how the concentration of antihistamine in the body varies over time.*

When Varying Individual Parameters, the concentration over time in different systems can be observed. These coefficients represent relative interaction areas, due to the assumption of equal solubility between compartments. The coefficients are generally constant for a population, however there could be some amount of variance due to varying physiology. Due to the limited data available to determine these parameters, a series of tests was performed, in which each parameter was varied, to extract a graph with a similar half-life as given in literature.

Using patient mass, and other similar characteristics such as body fat percentage, the ideal therapeutic dose can be experimentally determined. The ideal therapeutic dose is the dosage of the antihistamine that will last as long as possible, whilst not exceeding a designated concentration, which can be determined using the four compartment mass model.

This model has a several challenges that impose major limitations, which would require experimental values to negate, or further limitations to properly understand. One of these challenges is the necessity of assuming equal solubility in all necessary tissues, and negligible solubility in muscle and bone tissues. The other challenge is determining the rates of transmission coefficients, and solubility, to create a more accurate model. The current model is an approximation from the half-life in the body, and the digestive tract absorption, in the four-compartment model. This model is lightweight, with the majority of the computational time resulting from the creation of the output graphs for analysis. Though this lightweight computational model is effective for dosing based on known parameters, it has the ability to improve with access to better information on the transfer between compartments, and accounting for diffusion into each part of the body from the bloodstream. However, these additions may not be necessarily as important, as the immediate decay transfer does already simulate this.

6 Conclusion

Therefore, it is evident the model succeeds fairly well in modelling the system whilst being simple enough to produce relatively quick results. The ease of alteration of the parameters, and use of series of estimates to time the therapeutic effects has useful applications in the medical field in counteracting allergic reactions. Further research in the transfer of the drug between tissues, and half-lives of the drugs in different tissues is required to further the accuracy of the model. However, as it currently is, the model is effective enough to be used to fine tune an antihistamine dosage to a person's specific needs.

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8 Appendix

8.1 Complete Heun's Method Equations for 3.4.1

$$C_{b,n+1} = \frac{[f(\widetilde{C_{b,n+1}}, \widetilde{C_{f,n+1}}, \widetilde{C_{r,n+1}}) + f(C_{b,n}, C_{f,n}, C_{r,n})]\Delta t}{2} + C_{b,n} \quad (25)$$

$$C_{f,n+1} = \frac{[g(\widetilde{C_{b,n+1}}, \widetilde{C_{f,n+1}}) + g(C_{b,n}, C_{f,n})]\Delta t}{2} + C_{f,n} \quad (26)$$

$$C_{r,n+1} = \frac{[h(\widetilde{C_{b,n+1}}, \widetilde{C_{r,n+1}}) + h(C_{b,n}, C_{r,n})]\Delta t}{2} + C_{r,n} \quad (27)$$

where $\widetilde{C_{b,n+1}} = f(C_b, C_f, C_r)\Delta t + C_{b,n}$, $\widetilde{C_{f,n+1}} = g(C_b, C_f)\Delta t + C_{f,n}$, $\widetilde{C_{r,n+1}} = h(C_b, C_r)\Delta t + C_{r,n}$

8.2 Complete Runge-Kutta (4) Equations for 3.4.2

Initial slopes:

$$F_1 = f(C_{b,n}, C_{f,n}, C_{r,n}), \quad G_1 = g(C_{b,n}, C_{f,n}), \quad H_1 = h(C_{b,n}, C_{r,n}) \quad (28)$$

Halfway slopes:

$$F_2 = f(C_{b,n} + \frac{\Delta t}{2}F_1, C_{f,n} + \frac{\Delta t}{2}G_1, C_{r,n} + \frac{\Delta t}{2}H_1) \quad (29)$$

$$G_2 = g(C_{b,n} + \frac{\Delta t}{2}F_1, C_{f,n} + \frac{\Delta t}{2}G_1) \quad (30)$$

$$H_2 = h(C_{b,n} + \frac{\Delta t}{2}F_1, C_{r,n} + \frac{\Delta t}{2}H_1) \quad (31)$$

Readjusted Halfway slopes:

$$F_3 = f(C_{b,n} + \frac{\Delta t}{2}F_2, C_{f,n} + \frac{\Delta t}{2}G_2, C_{r,n} + \frac{\Delta t}{2}H_2) \quad (32)$$

$$G_3 = g(C_{b,n} + \frac{\Delta t}{2}F_2, C_{f,n} + \frac{\Delta t}{2}G_2) \quad (33)$$

$$H_3 = h(C_{b,n} + \frac{\Delta t}{2}F_2, C_{r,n} + \frac{\Delta t}{2}H_2) \quad (34)$$

End slopes:

$$F_4 = f(C_{b,n} + \Delta t F_3, C_{f,n} + \Delta t G_3, C_{r,n} + \Delta t H_3) \quad (35)$$

$$G_4 = g(C_{b,n} + \Delta t F_3, C_{f,n} + \Delta t G_3) \quad (36)$$

$$H_4 = h(C_{b,n} + \Delta t F_3, C_{r,n} + \Delta t H_3) \quad (37)$$

Then the value at the next time step is given by taking a weighted average of different slopes,

$$C_{b,n+1} = C_{b,n} + \frac{\Delta t}{6}(F_1 + 2F_2 + 2F_3 + F_4), \quad (38)$$

$$C_{f,n+1} = C_{f,n} + \frac{\Delta t}{6}(G_1 + 2G_2 + 2G_3 + G_4), \quad (39)$$

$$C_{r,n+1} = C_{r,n} + \frac{\Delta t}{6}(H_1 + 2H_2 + 2H_3 + H_4), \quad (40)$$

8.3 Complete Heun's Method Equations for 4.4.1

$$C_{b,n+1} = \frac{[f(\widetilde{C_{b,n+1}}, \widetilde{C_{f,n+1}}, \widetilde{C_{r,n+1}}, \widetilde{C_{d,n+1}}) + f(C_{b,n}, C_{f,n}, C_{r,n}, C_{d,n})]\Delta t}{2} + C_{b,n} \quad (41)$$

$$C_{f,n+1} = \frac{[g(\widetilde{C_{b,n+1}}, \widetilde{C_{f,n+1}}) + g(C_{b,n}, C_{f,n})]\Delta t}{2} + C_{f,n} \quad (42)$$

$$C_{r,n+1} = \frac{[h(\widetilde{C_{b,n+1}}, \widetilde{C_{r,n+1}}) + h(C_{b,n}, C_{r,n})]\Delta t}{2} + C_{r,n} \quad (43)$$

$$C_{d,n+1} = \frac{[m(\widetilde{C_{d,n+1}}) + m(C_{d,n})]\Delta t}{2} + C_{d,n} \quad (44)$$

$$\text{where } \widetilde{C_{b,n+1}} = f(C_b, C_f, C_r, C_d)\Delta t + C_{b,n}, \quad \widetilde{C_{f,n+1}} = g(C_b, C_f)\Delta t + C_{f,n}$$

$$\widetilde{C_{r,n+1}} = h(C_b, C_r)\Delta t + C_{r,n}, \quad \widetilde{C_{d,n+1}} = m(C_d)\Delta t + C_{d,n}$$

8.4 Complete Runge-Kutta (4) Equations for 4.4.2

Initial slopes:

$$F_1 = f(C_{b,n}, C_{f,n}, C_{r,n}, C_{d,n}), \quad G_1 = g(C_{b,n}, C_{f,n}), \quad H_1 = h(C_{b,n}, C_{r,n}), \quad M_1 = m(C_{d,n}) \quad (45)$$

Halfway slopes:

$$F_2 = f\left(C_{b,n} + \frac{\Delta t}{2}F_1, C_{f,n} + \frac{\Delta t}{2}G_1, C_{r,n} + \frac{\Delta t}{2}H_1, C_{d,n} + \frac{\Delta t}{2}M_1\right) \quad (46)$$

$$G_2 = g\left(C_{b,n} + \frac{\Delta t}{2}F_1, C_{f,n} + \frac{\Delta t}{2}G_1\right) \quad (47)$$

$$H_2 = h\left(C_{b,n} + \frac{\Delta t}{2}F_1, C_{r,n} + \frac{\Delta t}{2}H_1\right) \quad (48)$$

$$M_2 = m\left(C_{d,n} + \frac{\Delta t}{2}M_1\right) \quad (49)$$

Readjusted Halfway slopes:

$$F_3 = f \left(C_{b,n} + \frac{\Delta t}{2} F_2, C_{f,n} + \frac{\Delta t}{2} G_2, C_{r,n} + \frac{\Delta t}{2} H_2, C_{d,n} + \frac{\Delta t}{2} M_2 \right) \quad (50)$$

$$G_3 = g \left(C_{b,n} + \frac{\Delta t}{2} F_3, C_{f,n} + \frac{\Delta t}{2} G_3 \right) \quad (51)$$

$$H_3 = h \left(C_{b,n} + \frac{\Delta t}{2} F_2, C_{r,n} + \frac{\Delta t}{2} H_2 \right) \quad (52)$$

$$M_3 = m \left(C_{d,n} + \frac{\Delta t}{2} M_2 \right) \quad (53)$$

End slopes:

$$F_4 = f(C_{b,n} + \Delta t F_3, C_{f,n} + \Delta t G_3, C_{r,n} + \Delta t H_3, C_{d,n} + \Delta t M_3) \quad (54)$$

$$G_4 = g(C_{b,n} + \Delta t F_3, C_{f,n} + \Delta t G_3) \quad (55)$$

$$H_4 = h(C_{b,n} + \Delta t F_3, C_{r,n} + \Delta t H_3) \quad (56)$$

$$M_4 = m(C_{d,n} + \Delta t M_3) \quad (57)$$

Then the value at the next time step is given by taking a weighted average of different slopes,

$$C_{b,n+1} = C_{b,n} + \frac{\Delta t}{6} (F_1 + 2F_2 + 2F_3 + F_4) \quad (58)$$

$$C_{f,n+1} = C_{f,n} + \frac{\Delta t}{6} (G_1 + 2G_2 + 2G_3 + G_4) \quad (59)$$

$$C_{r,n+1} = C_{r,n} + \frac{\Delta t}{6} (H_1 + 2H_2 + 2H_3 + H_4) \quad (60)$$

$$C_{d,n+1} = C_{d,n} + \frac{\Delta t}{6} (M_1 + 2M_2 + 2M_3 + M_4) \quad (61)$$